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# Infrared thermal imaging and Doppler vessel pressurization ultrasonography to detect lower extremity deep vein thrombosis: diagnostic accuracy study

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**Short title：Infrared thermal imaging detect DVT**

## ABSTRACT

**Background and objective:** Infrared thermal imaging (IRTI) is a new technique for detecting deep vein thrombosis (DVT) based on DVT's infrared presentation and distribution characteristics (PDCs). A method that is singularly sensitive to DVT is needed. We therefore enrolled 157 subjects with suspected lower extremity DVT in a double-blind, controlled clinical trial using IRTI and Doppler compression ultrasonography (CPUS) to verify the clinical value of IRTI.

**Methods:** An IRTI system could precisely measure and store real-time thermal images. We conducted a double-blind, controlled clinical study using IRTI and detection on 157 patients with suspected DVTs, evaluating the following parameters: sensitivity, specificity, positive prescreening, negative prescreening, false positivity, false negativity, and diagnostic accordance of IRTI with CPUS for detecting DVT.

**Results:** Of 140 subjects who underwent screening both IRTI and CPUS detect were included for analyses. According to their IRTI DVT's presentation and distribution characteristics, patients were divided into IRTI- negative (n = 59) and IRTI-positive

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(including suspicious IRTI positive) (n = 81) groups. CPUS identified 80 DVT-negative and 60 DVT-positive patients. The sensitivity of IRTI for detecting DVT was 88.33%, specificity 65.00%, false-positive diagnosis 11.67%, false-negative diagnosis 35.00%, positive prescreening 65.43%, negative prescreening 88.14%, diagnostic accordance rate 75.00%. IRTI results accorded with CPUS results ($P$=0.001) except for the positivity incidence ($\chi^2 = 39.997$, $P < 0.001$).

**Conclusions:** IRTI could be used to supplement CPUS detection for detecting DVTs and adjunctive diagnostic screening. .

**Keywords** compression ultrasonography (CPUS), deep vein thrombosis (DVT), diagnostic accuracy study, infrared thermal imaging (IRTI), lower limb.

**SUMMARY AT A GLANCE:**

Infrared thermal imaging (IRTI) is a new, simple, noncontact, non-radiation, harmless tool. We have successfully verified that IRTI is an adjunctive diagnostic screening measure in detecting deep vein thrombosis (DVT) for early diagnosis. The purpose of this study is to certify the sensitivity and specificity of IRTI in detecting DVT.

## INTRODUCTION

Pulmonary thromboembolism (PTE) and deep vein thrombosis (DVT) of the lower limbs belong to the overall group of venous thromboembolisms (VTEs). About

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90% of PTEs come from DVTs. Among the more than 1 million DVT sufferers with a confirmed diagnosis, about 25% die from a PTE, and as many as 50% had already developed their PTE, which appeared as the initial symptom.[1] In the United States, 0.3 million people are seen for new consultations regarding DVTs annually, but the diagnosis cannot be confirmed in most of them because of the lack of well-defined symptoms. DVTs might lead to a fatal cardiovascular incident. DVTs at autopsy were found to be the cause of death in 72% of those who died with various diseases.[2]

The progress of a developing DVT is comparatively obscure clinically and without specificity. For about 80% of DVT sufferers, especially those whose thrombosis are subclinical, it is difficult to confirm their presence.[3] The d-dimer test is an important means of detecting DVTs, but it can only be used to exclude those who do *not* have a DVT, because operations, tumor or inflammation can lead to increase the value, which made its specificity was poor. Phlebography is the gold standard of DVT diagnosis, but it is invasive, costly, and causes the patient discomfort. Also, 1% of angiography evaluations are associated with complications.

We therefore propose a simple, noncontact, non-radiation, harmless, adjunctive diagnostic screening measure for patients highly suspected of having a DVT and those who do not have a DVT but are at high risk of developing one.[4] At present, Doppler compression ultrasonography (CPUS) is the technique of choice for harmless evaluation of a possible DVT[1] because it offers such advantages as high specificity, little harm, and no exposure to radiation. It can accurately confirm or exclude a DVT. Its sensitivity and specificity for diagnosing a proximal DVT are as high as 100% and

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98%, respectively. For a distal DVT, these parameters are 94% and 75%, respectively.[1,5,6] CPUS, however, is time-consuming and a doctor highly qualified in vessel detection with CPUS must be available as detection sensitivity is closely related to the instrument's precision and the knowledge and experience of the physician conducting the procedure.[5,7]

Infrared thermal imaging (IRTI) is a new detection tool with harmless, noninvasive, noncontact advantages. In previous studies, we used IRTI in a rabbit femoral vein thrombosis model and in a clinical trial to define DVT. That trial confirmed that IRTI detects the presentation and distribution characteristics (PDCs) of limb DVTs. Also, IRTI detects abnormal skin temperature of the lower limbs, which is significant for detecting arterial abnormalities[7,8] and DVTs.[9,10]

Based on the ability of IRTI to detect a DVT's PDC, patients suspected of having a DVT who were at high risk clinically and those with symptoms or signs of lower limb soreness or swelling were enrolled in a double-blind controlled clinical trial of IRTI and CPUS to determine such indexes as the sensitivity, specificity, negative prescreening, positive prescreening, false positivity, false negativity, and diagnostic accordance rate of IRTI with CPUS for detecting DVTs. We hypothesized that IRTI could be used as the standard for DVT detection and consultation. In doing this, we successfully verified that IRTI is an advanced detection technique that is highly sensitive to DVTs. It is a convenient, simple test that causes no harm and is easy to perform. It could be used as an adjunctive diagnostic screening test for early diagnosis.

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## METHODS

### Study population

A total of 157 participants comprised the study group. Each participant met the following inclusion criteria: They volunteered, and they were clinically suspected to have a DVT with the following clinical manifestations: lower limb with general edema, swelling, feeling of heaviness or soreness, pain or tenderness, skin pyrexia, pigmentation, and superficial varicose veins. Alternatively, the patients were at high risk of DVT clinically, mainly patients with a tumor, those who were postoperative, who had chronic obstructive pulmonary disease, were pregnant, or who had already experienced a PTE or DVT.

Ethics approval was obtained from Ethics Committee of First Affiliated Hospital of Guangzhou Medical University. Informed consent was signed by all patients.

### Procedures

The study was designed to be a diagnostic accuracy study —that is, the doctors conducting CPUS and IRTI did not know each other's results. Using CPUS detection as the comparison, we determined the sensitivity, specificity, positive prescreening, negative prescreening, false positivity, false negativity, and diagnostic accordance rate of IRTI with DPUS in regard to DVT.

All subjects were divided into DVT-negative and DVT-positive groups based on their CPSU results.

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The IRTI parameters included 320 × 240 pixels and a non-cooled focus plane array (ATIR-M301; AG Digital Technology Corp, Taiwan, China). They also included a spectral response of 8–14 μm, spatial resolution 1.3 mrad, temperature resolution 0.05°C, and acquisition speed 30 frames / s. The IRTI detection room was strictly temperature-controlled at ambient temperature (23°C±2°C). The relative humidity was about 60%, without air convection or direct sunlight radiation, which is technically required to control errors arising from the surroundings.

Before the IRTI examination, all subjects removed their clothing from both lower limbs and rested quietly for 10–15 min until emotionally stable and skin temperature had adapted to the room's temperature. These steps were necessary because clothing interferes with the body's infrared radiation, and activity increases infrared radiation. The infrared camera was turned on approximately 30 min prior to the onset of testing. Thus, all possible errors deriving from the device and the participants were minimized and kept stable.

The IRTI capture camera was positioned 2–3 m away from the subject. IRTI was performed on the lower extremities to capture both anterior and posterior standing views. For all participants, the knee or popliteal fossa was set as the center of the visual capture field, which covered the entire lower extremity. This IRTI system evaluated and analyzed changes in body surface temperature according to the IRTI-detected presentation and distribution characteristics, with different colors and temperature values. The "DVT limb" was that in which the DVT was located. The criteria for a negative versus a positive IRTI were as follows. The IRTI presentation

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and distribution characteristics in normal adult limbs was typically bilaterally symmetrical, which named negative IRTI. However, the positive IRTI was different, in “DVT limb”, specific color-temperature responses showed that the IRTI presentation and distribution characteristics on the DVT side was stronger (higher temperature) than on the control side, and the IRTI presentation and distribution characteristics in the regions of interest (ROI) on the DVT-affected side was conspicuously abnormal, showing an asymmetrical heat zone. Results indicated that the presentation and distribution characteristics of the body surface temperature in the DVT limb were higher (said to be IRTI-positive, Figure 1) than those in the other limb (said to be IRTI-negative, Figure 2). There were 17 subjects excluded from the total of 157 participants at last because of 8 persons did not meeting inclusion criteria, and 9 persons did not have CPUS measurement. The 140 subjects were divided into two groups: IRTI-positive and IRTI-negative.

**Statistical analysis**

Statistical analysis was performed using SPSS (PROC MIXED in SPSS 16.0 evaluation version; SPSS Inc., Chicago, IL, USA). The estimation of the accuracy was described by Sensitivity, specificity, positive predictive value, negative predictive value and accuracy, and using positive predictive and negative predictive values to instead of prescreening in this research. The difference between IRTI and CPUS was performed with Chi-square test, and the consistency between IRTI and CPUS was

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performed with McNemar test. $P < 0.05$ and $P < 0.01$ were considered significant and highly significant, respectively.

**RESULTS**

Of 140 subjects, aged 24–88 years (average 66.47 ± 14.77 years) with a male/female ratio of 83/57, who underwent screening both IRTI and CPUS detect were included for analyses (Figure 3). The patients' characteristics including underlying diseases or risk factors, symptoms and signs were summarized in table 1.

DVT-negative and DVT-positive were confirmed by CUPS evaluation and screening principles. Based on the CPUS results, 80 patients were judged DVT-negative and 60 patients were DVT-positive.

According to the IRTI evaluation of the DVT's presentation and distribution characteristics, all participants were divided into IRTI-negative and IRTI-positive groups (negative means that IRTI did not detect DVT presentation and distribution characteristics, and positive means that IRTI confirmed the presence of a DVT, including those inconclusive cases which had been named "suspiciously" IRTI-positive). Of all of subjects, 59 were found to be IRTI-negative and 81 IRTI-positive. The IRTI and CPUS results are shown in Table 2.

Thus, overall, 53 participants who were IRTI-positive were in the 60 CPUS-DVT-positive group, and 52 who were IRTI-negative were in the 80 CPUS-normal group. The sensitivity, specificity, positive prescreening, negative prescreening, and diagnostic accordance rate of IRTI with CPUS for detecting DVTs

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were 88.33%, 65.00%, 65.43%, 88.14%, and 75.00%, respectively. The two detection methods were different ($P = 0.001$). The only exception was the positivity incidence of IRTI, which was not in accord with that found by CPUS ($\chi^2 = 39.997$, $P < 0.001$) (Table 2).

## DISCUSSION

The pathological mechanism of DVTs is multifactorial. The most commonly recognized high-risk factors include surgery, presence of a tumor, trauma, less activity or bedridden, inactive because of illness, postoperative status, and taking drugs that result in increased blood viscosity or slow blood flow. It is therefore important to screen for these risk factors to detect DVT at an early stage.

Thermography—imaging the temperature distribution in the body— can also be used for diagnosing certain diseases, particularly in their early stage.[11] Generally, thermography can be achieved in two ways: (1) contact thermography with liquid crystal contact screens (LCCT) used to cover two-dimensional parts of the body; and (2) noncontact thermography (IRTI), which uses an infrared non-cooled camera to capture infrared radiation emitted from the body surface. The latter records and measures body surface temperatures.[11] The first use of thermography for diagnosing DVT was suggested in 1972 and further established during 1973–1974.[12] Since then, several studies on LCCT detection and diagnosis of DVT have confirmed that thermography is a highly effective screening method for DVT, with thermography yielded sensitivities of 55–97%, specificities of 62–85%, negative prediction values of 94.0–96.5%, positive prediction value of 67.7%, and 90% agreement between LCCT

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and phlebography.[12-17] To date, however, there has been no report on screening for DVT with IRTI.

In this research, the results showed that both CPUS and IRTI detection could be applied to DVTs or patients with high risk factors. The two techniques are different ($P$ =0.001). According to the established diagnostic criteria for IRTI, DVT sensitivity and specificity were 88.33% and 65.00%, respectively. False positivity and false negativity were 11.67% and 35.00% respectively. Positive prescreening and negative prescreening was 65.43% and 88.14%, respectively. The diagnostic accordance rate of IRTI and CPUS for detecting DVT was 75%, which confirmed that the DVT diagnostic criteria we preliminarily established are feasible, with important application value. However, the disagreement regarding the positivity incidence from CPUS and IRTI (v = 39.997, $P < 0.001$) showed that CPUS and IRTI are not interchangeable. Thus, IRTI results correlated well to CPUS results but one cannot replace the other. IRTI can thus be used to supplement CPUS detection.

IRTI is a functional thermal imaging detection system, with the help of an infrared non-cooled camera, it captures, records, and measures the infrared radiation being emitted from the human body's surface and can detect its surface temperature.[19-21] This technique was initially used for nighttime detection by the US army. Because of the development of infrared non-refrigerated lenses, high-resolution digitization picture, and infrared thermal imaging analytic software—and because of its advantages of low cost, noninvasiveness, and noncontact—IRTI has been gradually

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applied in many medical fields.[20, 21] However, at present, like other imaging detection methods such as magnetic resonance imaging, computed tomography, and plain radiography, CPUS is directed at evaluating anatomical structure. Despite the fact that these imaging detection techniques can well look through the human body and provide the information about abnormal structure and pathology in multi-angular and all-dimensional situations, they are limited in that they can only indicate differences in morphological structure of the tissue. They cannot fully disclose the organizational function of the human body. CPUS, in contrast, is able to observe the structural state of the human body.

The detection ability of IRTI is based on physiology. Here, the functional properties of the human body can be observed by infrared heat (of varying degree) radiating from the human body that accurately reflects the features and regularity at the cellular metabolic level. Considering the fact that functional change is controlled by the dynamic process of human physiology, no matter how sophisticated the human body is, all of the heat in the inner body transfers to the surface. Heat distribution is the overlay of many simple approximate distributions of the body. Hence, dynamic observation of human body function is more premonitory than static, which explains why the two detection techniques have different positivity rates. In the early stage of the disease, the functional change is earlier than structure change, so the primary reason is that IRTI can make earlier discoveries. Disease is more easily detected when the bloodstream appears slow—but not at DVT status. CPUS cannot provide a definitive diagnosis until the slow blood flow progresses to form a DVT.

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It would be more perfect and effective when CPUS structure imaging combined with IRTI functional imaging, however, in this research, we found there were several patients, they were IRTI positive, whereas CPUS negative. The main reasons were ① No vascular embolization, but exiting early functional changes, such as blood flow slowing down, higher deep vein blood flow resistance. ② The embolus vessel was recanalization, the thrombus embolism was dissolved and no emboli in the vessels. They would be the reasons why the IRTI failure rate were clearly too high in this research.

Thus, IRTI could be used for detecting DVTs and adjunctive diagnostic screening, and the effective combination of IRTI and CPUS detection would be of great value for early-stage detection and definitive diagnosis of DVT.

However, this study has several limitations. Many factors had interfered with the IRTI analysis, such as some inconclusive cases, that is “suspiciously IRTI-positive”, had been included in IRTI positive, which inducing directly the clearly high IRTI failure rate. There also had deficiencies in study design such as including 5 follow visited patients, low sample size. Because of the research participants including patients with stroke, injured, and skin wounds, which leaded to certain of heterogeneities.

**Transparency:**

**Declaration of funding:**

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**Declaration of financial interests:**

All authors declare that they have no relationships with or financial interests in any commercial companies related to this study and article. All authors declare that they have no conflicts of interest to report.

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**Ethics:**

This research was approved by the Ethics Committee of The First Affiliated Hospital of Guangzhou Medical University. All patients gave their informed consents.

**Clinical Trial Registration:**

ClinicalTrials.gov; No.: NCT02552420; URL: www.clinicaltrials.gov.

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**Figure 1 DVT-IRTI-positive**

The PDC of IRTI was positive in DVT patients, showing the body surface temperature in the DVT limb were higher

**Figure 2 DVT-IRTI-negative**

The PDC of IRTI was negative in non-DVT patients , showing the body surface temperature in the limb were normal

**Figure 3 Subject recruitment flow chart**

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**Table 1 The patients' underlying diseases or risk factors, symptoms and signs**

| | | Number of patients |
|---|---|---|
| Underlying diseases / Risk factors | VTE (PTE or/and DVT) | 57 (40.71%) |
| | Chronic obstructive pulmonary disease | 37 (26.43%) |
| | Malignancy | 35 (25%) |
| | Postoperation | 31 (22.14%) |
| | Trauma, wound, fracture, contusion | 10 (7.14%) |
| | Pregnancy | 3 (2.14%) |
| | Immobilization or obviously less activities | 93 (66.43%) |
| | High blood pressure/lipids, diabetes | 69 (49.29%) |
| | Arteriosclerosis, heart diseases | 47 (33.57%) |
| | Cerebrovascular diseases | 12 (8.57%) |
| | Oral contraceptives | 1(0.71%) |
| Symptoms / Signs | Lower extremity edema or swelling | 67 (47.86%) |
| | lower extremity varicose veins | 54 (38.57%) |
| | Faintness, lacking strength | 54 (38.57%) |
| | feeling of heaviness, pain, soreness | 41 (29.29%) |

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**Table 2 Distribution of DVT patients according to performance of IRTI and CPUS: double-blind, controlled screening results**

| | DVT (CPUS pos) | No DVT (CPUS neg) | Total |
|---|---|---|---|
| IRTI pos | 53 (37.86%)[(a)] | 28 (20%) [(b)] | 81 (57.86%) |
| IRTI neg | 7 (5%) [(c)] | 52 (37.14%)[(d)] | 59 (42.14%) |
| Total | 60 (42.86%) | 80 (57.14%) | 140 (100%) |

Sensitivity: a/(a+c)=88.33%, 95% CI (77.4, 95.2 ); specificity: d/(b+d)=65%, 95% CI (53.5, 75.3); positive predictive value: a/(a+b)=65.43%; negative predictive value: d/(c+d)=88.14%; accuracy: (a+d)/(a+b+c+d)=75%

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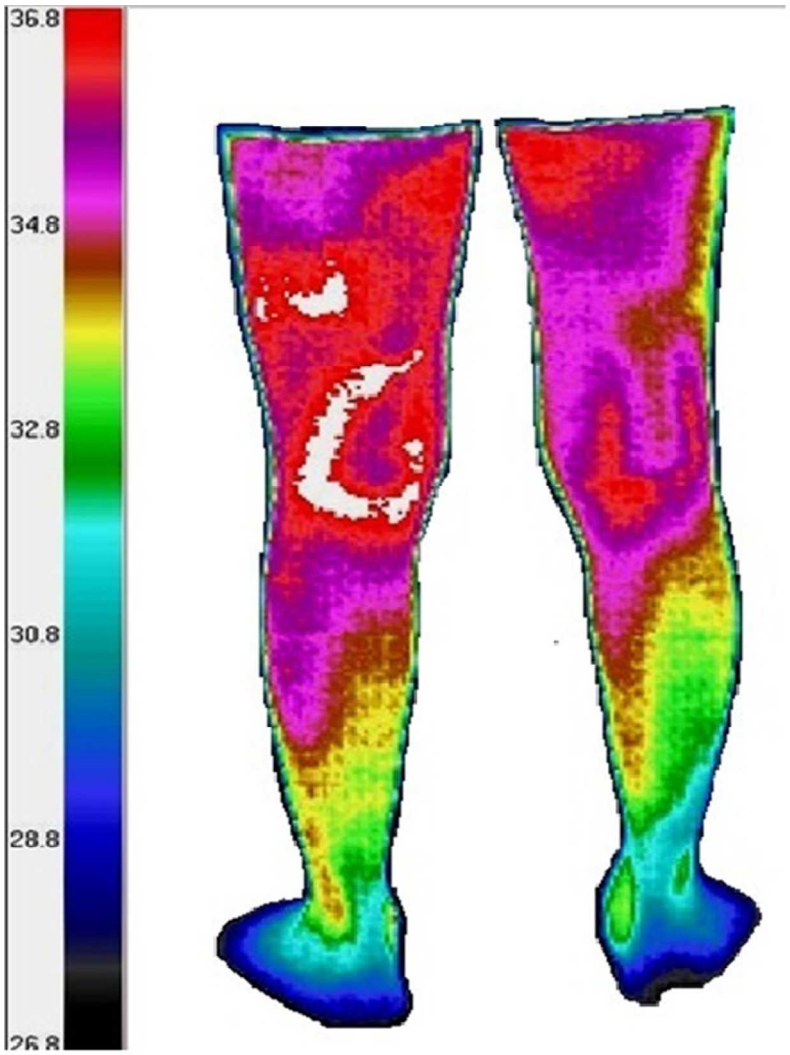



**Figure 1. DVT-IRTI-positive**

The PDC of IRTI was positive in DVT patients, showing the body surface temperature in the DVT limb were higher

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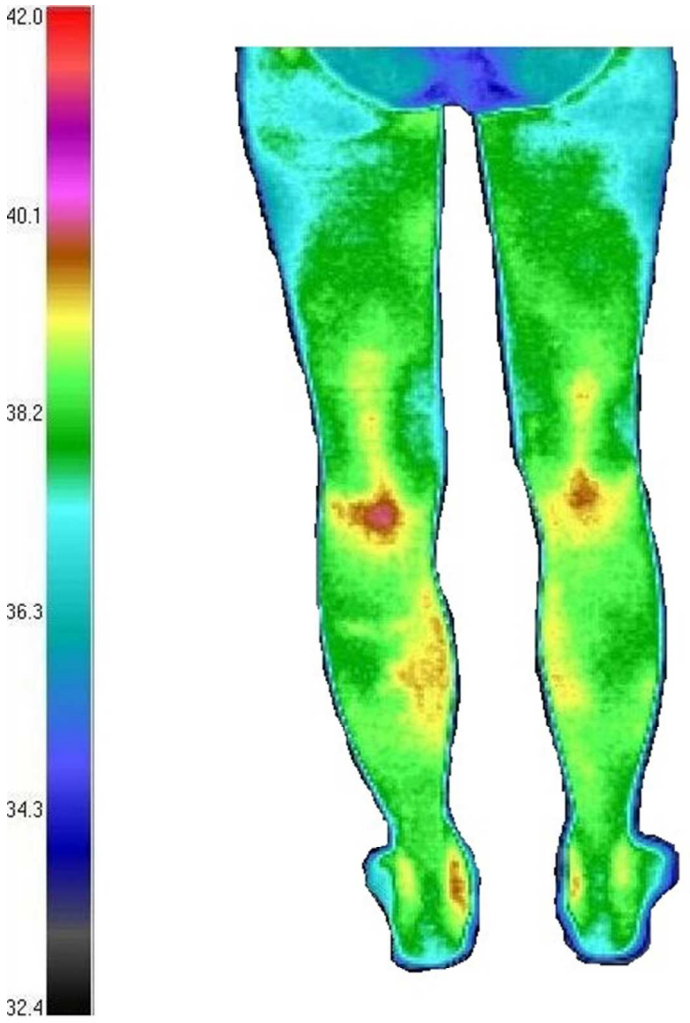



**Figure 2. DVT-IRTI-negative**

The PDC of IRTI was negative in non-DVT patients , showing the body surface temperature in the limb were normal

139x200mm (300 x 300 DPI)

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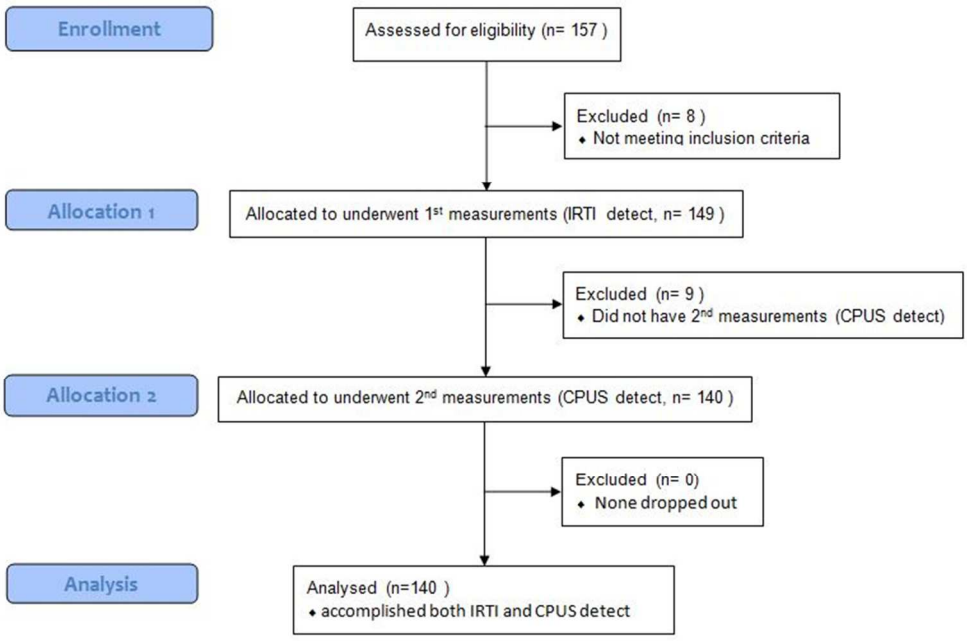



**Figure 3**. Subject recruitment flow chart.

IRTI= infrared thermal imaging. CPUS= doppler compression ultrasonography.

208x161mm (300 x 300 DPI)

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